Solve the sollwing System

E-Vex

$$\begin{cases} 2\ln x + \ln y = 4 \\ \ln x - 3\ln y = 9 \end{cases}$$

$$2\ln x + \ln y = 4$$

$$\ln(x^2 y) = 4$$

$$e^{\ln(x^2 y)} = e^4$$

$$x^2 y = e^4$$

$$\ln x - 3\ln y = 9$$

$$\ln\left(\frac{x}{y^3}\right) = 9$$

$$e^{\ln\left(\frac{x}{y^3}\right)} = e^9$$

$$\frac{x}{y^3} = e^9$$

$$\begin{cases} x^2 y = e^4 \\ \frac{x}{y^3} = e^9 \end{cases}$$

$$\boxed{x = y^3(e^9)}$$

$$\left(y^3\left(e^9\right)\right)^2 y = e^4$$

$$y^6 \cdot e^{18} \cdot y = e^4$$

$$y^7 \cdot e^{18} = e^4$$

$$y^7 = e^{-14} \Rightarrow \sqrt[7]{y^7} = \sqrt[7]{e^{-14}}$$

$$\boxed{y = e^{-2}}$$

then : $x = \left(e^{-2}\right)^3\left(e^9\right)$

$$x = e^{-6} e^9$$

$$\boxed{x = e^3}$$

$$S : \left\{\left(e^3, e^{-2}\right)\right\}$$

E-[signature]